Pakistan's EV and Battery Industrialisation

Policy, Market and Supply Chain Evidence for Chinese Investors

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Why This Report Exists

Almost everything written about Pakistan's electric vehicle sector over the past year has suffered from the same defect. It treats a ministerial press statement, a signed memorandum of understanding, a pitchbook target and a gazetted law as though they were the same category of fact. They are not. A gazetted law creates a right that a manufacturer can enforce and a lender can underwrite. A pitchbook target creates nothing except an impression. When these get blended together in a single narrative about Pakistan attracting billions in Chinese battery investment, the resulting picture is not merely optimistic, it is structurally misleading in ways that could cost a foreign investor a great deal of money.

I have written this report to separate those categories and keep them separate throughout. Every claim about policy in the pages that follow carries an explicit status, and I have tried to be consistent about the vocabulary. Notified means the instrument has completed its legislative or cabinet process and is in force today. In principle means a committee or steering group has reached agreement on the shape of something that has not yet been through formal approval. Draft or pitchbook means the number exists in a promotional or consultative document and carries no legal weight whatsoever. Where I do not know which of these applies, I say so rather than guessing, because a candid gap in the evidence is more useful to an investor than a confident invention.

The second thing this report tries to do is apply a single consistent test to every project it examines. If a facility imports finished electrochemical cells and builds them into modules and packs, that is assembly, which is genuine and valuable industrial activity but is not manufacturing in the sense the word carries when a government promises a battery industry. If a facility produces cells from active materials with real local process control, that is manufacturing. The distinction sounds pedantic until you notice how many Pakistani projects are described publicly using the second word while performing the first activity. I have applied the test to every named venture in section four, and the results are uncomfortable but, I think, accurate.

The third thing is arithmetic. Several of the numbers that circulate most freely in coverage of this sector fall apart when the base is examined. The most repeated statistic of the year, a sixty one per cent monthly jump in electric vehicle sales, describes a movement from thirty three units to fifty three units. The most repeated cost comparison, four thousand rupees against sixty five thousand, rests on one anecdote rather than any published tariff. Neither figure is fabricated, and both point in a direction that is genuinely real, but reproducing them without their denominators does a disservice to anyone trying to size this market seriously. Where I have found a more defensible calculation, I have shown the working so the reader can check it rather than take my word for it.

What follows is organised in six parts. Section one maps the policy architecture instrument by instrument, and it opens with a finding that surprised me during the research: Pakistan already has a notified electric vehicle policy, funded by a dedicated levy passed as an act of parliament, and its implementation record is considerably more informative about execution risk than any amount of analysis of the unfinished battery policy. Section two examines the market data. Section three examines the supply chain and the mineral base that Pakistani officials cite as justification for the entire localisation strategy. Section four is a named tracker of Chinese engagement, sorted by how much confidence each deserves. Section five sets Pakistan against Bangladesh and Vietnam, the two markets competing for the same pool of capital. Section six sets out what to watch.

1. The Policy Architecture

1.1 The Instruments That Actually Exist, and Which Ones Bind

Discussion of Pakistani electric vehicle policy tends to focus almost entirely on the National Lithium Ion Battery Manufacturing Policy, which remains unfinished, while overlooking the fact that Pakistan already has a notified electric vehicle policy that has been in force for a year. The New Energy Vehicle Policy 2025 to 2030 was approved by the federal cabinet on 26 August 2025. It is not a draft, not a consultation document and not a pitchbook. It is government policy with a legal funding mechanism attached, and anyone assessing Pakistan's seriousness about electrification should begin here rather than with the battery policy that has attracted more commentary.

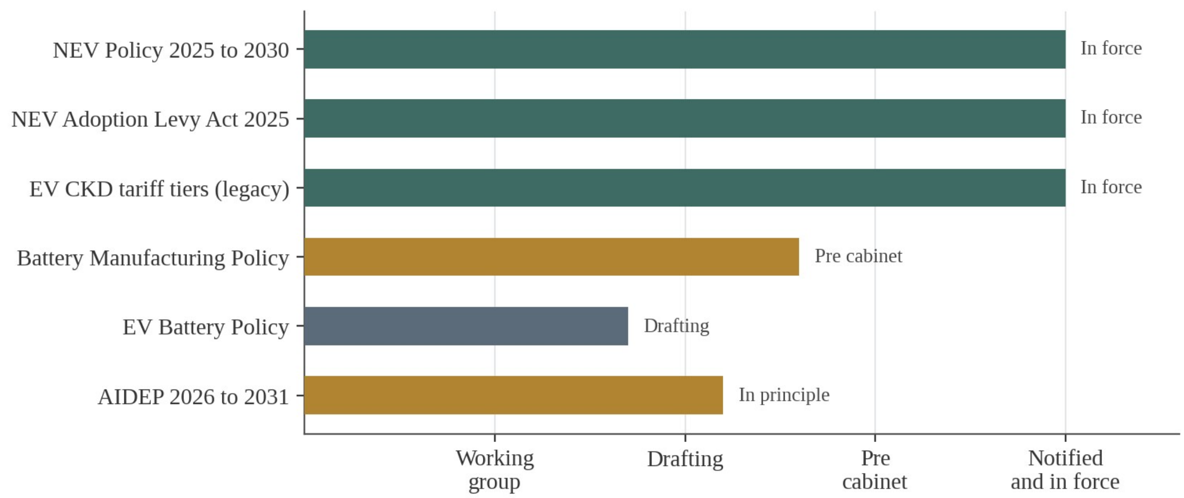
The policy sets a headline target of thirty per cent of new vehicle sales being electric by 2030, rising to fifty per cent by 2040 in the version approved by cabinet, with earlier drafts of the framework having proposed a more aggressive ninety per cent by 2040. That downward revision between draft and approval is itself a small but instructive illustration of the point this report keeps returning to, which is that targets circulating during consultation are routinely softened before they become policy. The instrument was developed over roughly a year of work involving more than sixty experts and institutions under a steering committee within the Ministry of Industries and Production established in September 2024, and it carries a governance structure that is unusually specific by the standards of Pakistani industrial policy, with monthly and quarterly review meetings by the steering committee and a performance audit by the Auditor General of Pakistan every six months.

What makes this policy genuinely different from the aspirational documents that surround it is that its money is real and legally sourced. The Pakistan Accelerated Vehicle Electrification programme, universally abbreviated PAVE, is funded through the New Energy Vehicle Adoption Levy Act 2025, which imposes a climate support levy of two rupees and fifty paisa per litre on petrol and diesel sold nationally. This is a revenue neutral design of a kind that development economists generally approve of, taxing the externality that policy is trying to reduce and recycling the proceeds into the substitute. It also means the subsidy programme does not compete against health, education or defence for a discretionary budget allocation each year, which in the Pakistani fiscal context is a considerable structural advantage and makes the funding stream far more durable than a line item that has to be defended annually.

The subsidy schedule underneath the policy is specific enough to be tested. Electric two wheelers receive up to sixty five thousand rupees, electric three wheelers up to four hundred thousand rupees, and four wheelers fifteen thousand rupees per kilowatt hour of battery capacity, alongside free registration for eligible vehicles. The initial allocation was nine billion rupees for the financial year, covering a target of roughly one hundred and sixteen thousand electric motorcycles and three thousand one hundred and seventy one rickshaws and loaders. The deliberate weighting toward two and three wheelers rather than passenger cars is the single most sensible feature of the entire framework, and it deserves more attention than it has received. Over ninety per cent of parts for two and three wheelers are already manufactured locally in Pakistan, which means subsidy money spent in this segment flows into an existing domestic supply chain rather than leaking straight back out as imports, and it targets the vehicle categories that account for the largest share of daily urban trips and fuel consumption.

Set against this notified and funded instrument, the other five policy documents governing the sector occupy weaker positions. The tariff structure for completely knocked down vehicle assembly is notified and in force but was inherited from an expired policy and now runs on extension. The National Lithium Ion Battery Manufacturing Policy sits at pre cabinet stage. The EV Battery Policy is still being drafted. The successor auto policy has reached in principle consensus at steering committee level but has not moved through the Ministry of Industries or the cabinet. Figure 1 shows the six instruments plotted against how far each has actually travelled, and the picture it produces is one an investor should sit with before reading anything else in this report, because the three instruments that bind are all on the demand and vehicle side, while every instrument that would govern battery manufacturing specifically remains unfinished.

Figure 1. Legal status of the six instruments governing the sector



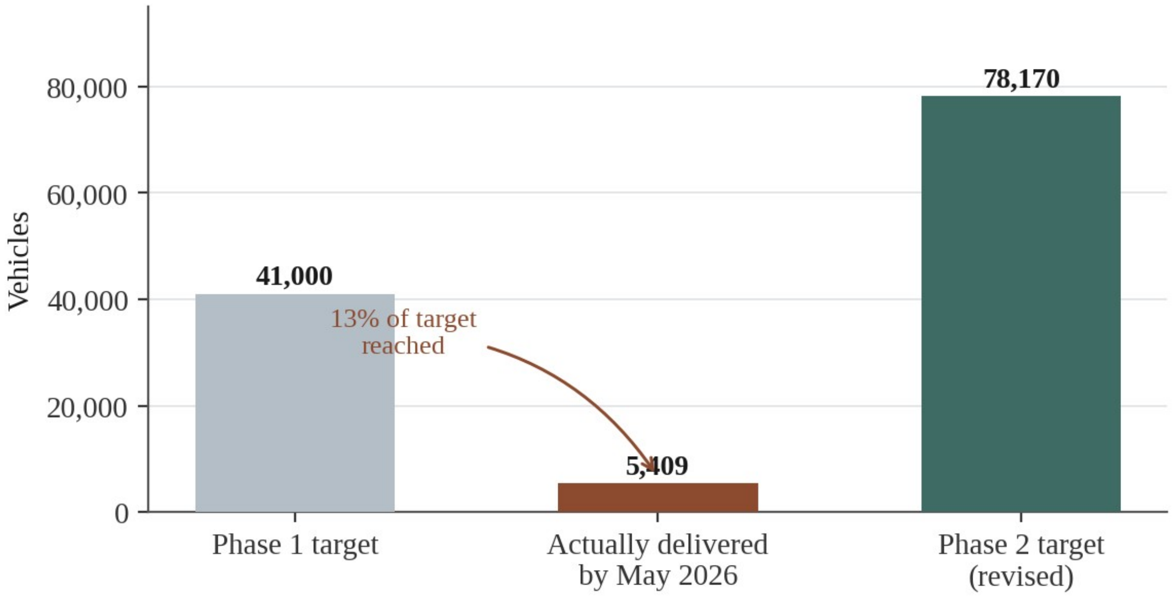
Only three of the six instruments governing this sector are currently enforceable, and none of the three governs battery manufacturing.

1.2 What PAVE Reveals About Execution Risk

The value of having a notified policy with a year of implementation behind it is that it produces something no amount of analysing draft documents can produce, which is evidence about whether Pakistani institutions can actually deliver what they announce. On this measure the record is sobering, and I would argue it is the most important single piece of evidence in this entire report for an investor trying to price execution risk, because it is the only genuine track record available.

The first phase of PAVE targeted forty one thousand vehicles. By the time the Economic Coordination Committee reviewed the programme in May 2026, five thousand four hundred and nine electric bikes and rickshaws had actually reached recipients. That is roughly thirteen per cent of the stated target delivered against a scheme that was fully funded, politically prioritised, publicly launched with considerable fanfare and administered by an established body in the Engineering Development Board. Figure 2 shows the shortfall against the revised second phase target for context.

Figure 2. PAVE subsidy scheme: target against actual delivery



The gap between announcement and delivery is the most reliable single indicator of execution risk in this sector.

What makes this shortfall genuinely instructive rather than simply disappointing is the diagnosis. The bottleneck was not citizen demand, which by all accounts substantially exceeded the available allocation, and it was not manufacturer capacity, since local two wheeler assemblers had ample ability to supply. It was bank side approval. The original scheme design required buyers to pay the full purchase price upfront and then claim reimbursement of the subsidy afterwards, a structure that pushed the entire financing burden onto applicants who were, almost by definition, the low income buyers least able to bridge it and least likely to secure quick bank facilities. The programme failed at the point where policy design met financial plumbing.

The government's response to that failure is, to its credit, an unusually direct fix rather than a rhetorical one. Phase two, approved by the Economic Coordination Committee in May 2026 with a nine billion rupee package, reverses the payment flow entirely. Buyers now pay the net price after subsidy deduction directly to the manufacturer, and the Engineering Development Board transfers the subsidy amount to the manufacturer through the State Bank of Pakistan once delivery and registration are verified. The allocation mechanism has also shifted from a computerised lucky draw to first come first served, removing a layer of delay and opacity. A revised self financing scheme for federal employees in grade sixteen and below requires a down payment of only ten thousand rupees for a bike. The revised phase two target has been set at seventy eight thousand one hundred and seventy vehicles.

For a Chinese investor, three separate lessons follow from this episode, and they generalise well beyond the two wheeler segment. The first is that Pakistani policy design has a demonstrated tendency to place financing friction on the party least equipped to absorb it, which is a risk worth checking for explicitly in any incentive scheme a project intends to rely on. The second, more encouraging, is that the system corrected itself within roughly nine months once the failure became visible, which is faster than many comparable jurisdictions manage and suggests the review architecture built into the policy is functioning rather than decorative. The third is that headline targets in Pakistani industrial policy should be discounted by a substantial and empirically grounded factor rather than taken at face value. If a fully funded, politically prioritised subsidy scheme delivered thirteen per cent of its first phase target, then the three thousand charging station target discussed in section two, and the localisation percentages discussed in section one point six, deserve to be read through the same lens.

1.3 The National Lithium Ion Battery Manufacturing Policy

The battery policy has a precisely datable origin, which is unusual enough in Pakistani policymaking to be worth noting on its own. A dedicated working group was constituted on 12 December 2025. Within weeks, a high level meeting chaired by Haroon Akhtar Khan, the Special Assistant to the Prime Minister on Industries and Production, had approved lithium iron phosphate as the chemistry to be prioritised for first phase localisation. By April 2026 the policy had been forwarded to the Ministry of Industries and Production and was awaiting consideration by the National Tariff Board, after which it would go to the Prime Minister and then to cabinet. As of the date of this report it had not completed that journey.

The chemistry decision deserves more credit than it has generally received, because it is the one part of this policy that reflects genuine strategic thinking rather than aspiration. Lithium iron phosphate cells are cheaper than the nickel manganese cobalt chemistry that dominates premium vehicles, considerably more thermally stable, which matters in a country with Pakistan's ambient temperatures and its uneven record on industrial safety enforcement, and better suited to the stationary storage applications that Pakistan's solar boom is generating demand for. Most importantly, the two elements that give the chemistry its name are the two raw materials Pakistan can most plausibly claim a domestic base in. Choosing a chemistry that matches your geology rather than the global premium standard is exactly the kind of decision a country in Pakistan's position should be making, and it stands in favourable contrast to jurisdictions that have chased the highest specification chemistry without any corresponding raw material logic.

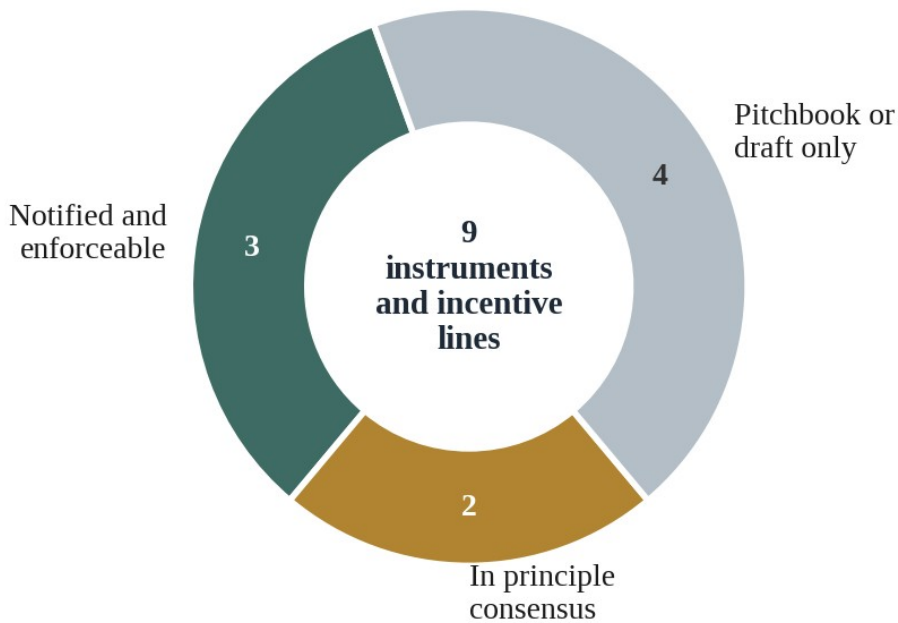
Around that chemistry decision, the policy is built on three stated pillars: phased localisation, tariff rationalisation and performance linked incentives. The localisation roadmap that has circulated alongside it targets seventy per cent domestic value addition in an initial implementation phase, rising to eighty per cent thereafter. These figures appear consistently across the reporting, and they are the numbers most frequently quoted back as though they were policy. They are not. They are targets described in an investment pitchbook and in ministerial briefings. Until the instrument clears the National Tariff Board and the cabinet, they describe an intention rather than an obligation a manufacturer can be held to or, more relevantly for an investor, a concession a manufacturer can rely on.

There is a further structural complication that most coverage misses entirely. The battery policy is not a single document awaiting one signature. It is an umbrella containing at least three work streams running on separate clocks. The localisation and tariff framework itself moved fastest. The

Pakistan Standards and Quality Control Authority testing, certification and recycling regime, which the policy assigns as a precondition, was still being developed as of mid August 2026. The broader Next Generation Energy Storage Policy that is meant to sit above both was, in the same reporting, described as still being finalised. An investor evaluating a battery project in Pakistan today is evaluating something whose eventual tariff treatment depends on how and when three separate tracks converge, and no public document specifies which takes precedence where they conflict.

Figure 4 summarises the practical consequence across the nine policy instruments and incentive lines this report has examined. Three are notified and enforceable. Two rest on in principle consensus. Four exist only in pitchbook or draft form. That ratio, rather than any individual headline target, is the honest summary of what an investor can currently build a financial model around.

Figure 4. What an investor can actually rely on today



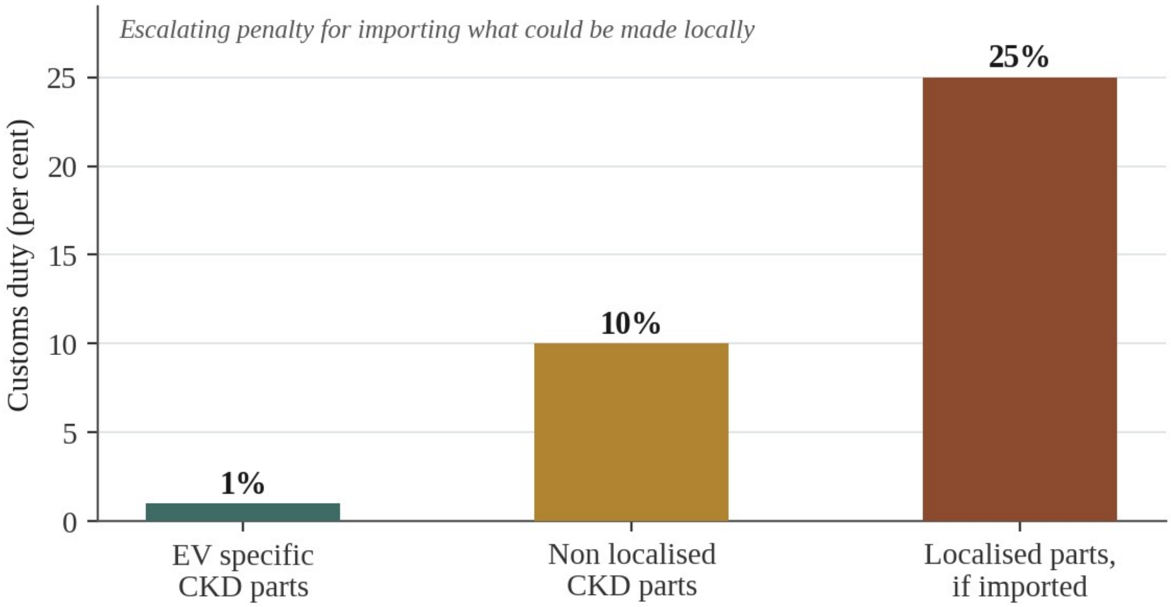
Four of the nine incentive instruments an investor might reasonably expect to rely on exist only in draft or promotional form.

1.4 The Tariff Schedule and What Is Genuinely Firm

Separating the enforceable from the announced matters most acutely on tariffs, because tariffs are where an industrial project's economics are actually decided. On the vehicle side, Pakistan already has a functioning notified structure, inherited from the expired auto policy and currently running on extension to June 2027. Electric vehicle specific components imported for completely knocked down assembly carry a duty of one per cent. Non localised completely knocked down parts, meaning components a manufacturer imports where a domestic supplier exists, carry ten per cent.

Parts that have been localised within an approved framework carry twenty five per cent if imported rather than sourced domestically.

Figure 3. Notified CKD tariff structure, in force to June 2027



The inverted structure is deliberate: the more a component could have been made locally, the more expensive importing it becomes.

That inversion is the cleverest feature of the existing framework and it is worth understanding precisely, because it is doing real work. Under a conventional tariff structure, a manufacturer pays less duty the more processed the imported item is, which rewards importing finished goods. Under this structure the penalty rises as a part moves onto the localised list, so a manufacturer who continues importing something Pakistani suppliers have begun making faces an escalating cost. The list of localised parts is updated through periodic revision of the relevant statutory regulatory order, which means the penalty perimeter expands as domestic capability grows. This is genuine industrial policy machinery rather than protection dressed up as strategy, and it is notified law rather than intention.

On the battery side specifically, none of the equivalent figures has that status. A zero per cent duty on imported cells through 2028, a twenty per cent federal excise duty on imported completely built battery units, a one per cent sales tax for local battery manufacturers and accelerated depreciation reported as high as ninety per cent all appear in official investment promotion material and ministerial briefings. They describe where the government wants to end up. None can currently be relied upon as an enforceable right, and a project whose financial model assumes a zero per cent cell import duty that has not been gazetted is carrying a policy risk quite separate from the ordinary commercial risk of operating in Pakistan.

One narrow concession does have notified status and is worth flagging precisely because of what its survival reveals about government preference. When the previous auto policy expired at the end of June 2026 without a successor in place, the government extended rather than allowed to lapse a one per cent sales tax concession specifically for locally assembled electric vehicles with battery capacity up to fifty kilowatt hours, and light commercial electric vehicles up to one hundred and fifty kilowatt hours, through June 2027. Incentives for hybrids were not so lucky. Sales tax on locally assembled hybrid and plug in hybrid vehicles above 1,400cc rose to twenty five per cent, replacing a concessional rate of eight point five per cent. Pure electric vehicles below a defined battery size kept their favourable treatment while larger hybrids lost theirs, which tells you where the government's revealed preference sits regardless of what any pitchbook says about the destination.

There is a further tariff development that cuts against the localisation narrative and deserves mention because it has generated real industry anger. A broader tariff rationalisation took effect at almost exactly the moment the old auto policy expired, and industry figures have argued publicly that the resulting structure is import biased in a way that rewards trading activity over domestic value addition. One senior industry finance executive put the criticism directly, noting that at a moment when Pakistan urgently needs industrialisation, foreign investment, exports and productive employment, a tariff structure tilted toward imports risks rewarding precisely the wrong behaviour. Whether that criticism is fully fair is contestable, since tariff rationalisation has independent macroeconomic justifications tied to Pakistan's commitments under its external financing arrangements, but the timing has left manufacturers without a coherent roadmap at exactly the point when several were making multi year capital commitments.

1.5 The EV Battery Policy: Intention Without a Text

A separate instrument, generally called the EV Battery Policy, is being developed alongside the battery manufacturing policy, and the two are conflated in press coverage often enough that untangling them is worthwhile. What can be stated with confidence is that the government has repeatedly and publicly committed to finalising a dedicated policy covering electric vehicle batteries and adjacent energy storage, that officials at Special Assistant level have described this as an active priority tied explicitly to the national energy security framework, and that lithium iron phosphate has been confirmed as the chemistry across both vehicle and stationary storage applications. Officials have also stated an intention to formally follow up on memoranda already signed by Chinese firms, which signals that the government views these agreements as a pipeline to be converted rather than a public relations exercise.

What remains unresolved is everything that would allow a project to be priced. The detailed incentive schedule has not been published in final form. The rollout timeline exists only as the word soon, which in the recent history of Pakistani industrial policy has meant anywhere from months to several years. Most consequentially for anyone structuring a transaction today, no public document specifies how the EV Battery Policy will interlock with the National Lithium Ion Battery Manufacturing Policy, which takes precedence where they overlap, or how conflicts between them will be resolved. As of mid August 2026 the Ministry was still developing the underlying battery standards while the energy storage policy above it was still being finalised, confirming that this is a live process rather than a text awaiting signature.

The right response to this is not to avoid the sector but to structure around the uncertainty rather than assume it away. A joint venture agreement that fixes a specific tariff or tax rate as a contractual term, on the assumption that the eventual policy will simply mirror the current pitchbook, is exposed to a risk neither party controls. The more defensible structure stages the investment against policy milestones. An initial tranche can proceed on the basis of what is already notified, covering pack assembly, battery management system integration and testing infrastructure, none of which depend on the unfinished cell level schedule. Later tranches, particularly anything committing to cell manufacturing or upstream material processing, should be conditioned on formal gazetting and should carry a defined renegotiation mechanism if the notified terms diverge materially from what was described during negotiation.

It is worth being explicit about what would count as genuine confirmation, so that future developments can be measured against a fixed bar rather than another round of statements. Three things: a gazette notification of the finalised policy, publication of the Pakistan Standards and Quality Control Authority battery safety and certification standards in final form, and a published tariff schedule specific to cells, cathode materials, anode materials and finished packs that supersedes the pitchbook figures. Until all three have occurred, this instrument should be classified as evidenced intent rather than commitment. That is not scepticism about Pakistan's seriousness. It is the same discipline any competent adviser would apply to a draft regulation in any jurisdiction, and Pakistan's automotive sector has a documented history of policy documents circulating in draft for years before notification, or lapsing entirely and being replaced by a differently structured successor.

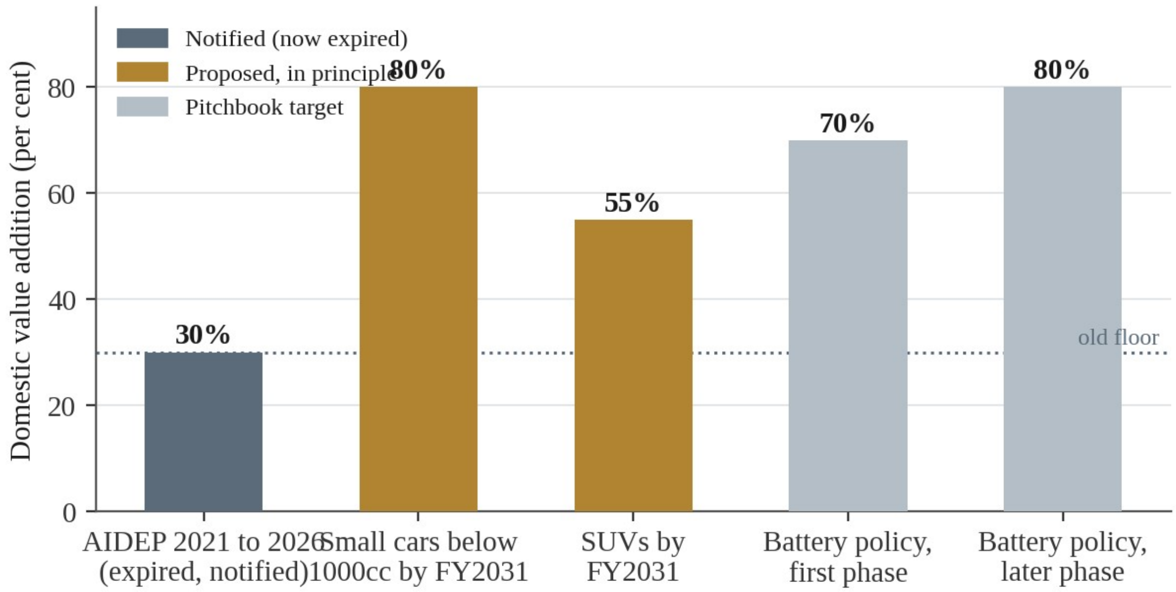
There is one asymmetry in this uncertainty that favours an early mover, and it deserves acknowledging rather than being buried under the caution. Because the incentive schedule is unfinished, it is also still open to influence. Pakistan's policy process for this instrument has explicitly involved industry consultation, with more than sixty stakeholders contributing to the parallel electric vehicle policy. A Chinese manufacturer with genuine capital at stake and a credible commitment to local value addition is in a materially better position to shape the final text than one arriving after gazettal, when the terms are fixed and the only remaining question is compliance. Investors who read regulatory uncertainty exclusively as risk sometimes miss that in a jurisdiction actively drafting, uncertainty is also access.

1.6 Local Content and the Successor Auto Policy

The Auto Industry Development and Export Policy 2021 to 2026 expired on 30 June 2026, leaving the sector without a governing framework while its successor remained under negotiation. Under the expired policy the local value addition requirement for components manufactured under the relevant concession scheme was fixed at thirty per cent, with anything below that threshold simply ineligible for concessional treatment. It also set an aspiration of localising one hundred per cent of motorcycle parts and seventy five per cent of car parts by 2026, an aspiration substantially met in the two wheeler segment, where local content now exceeds ninety per cent, and substantially missed in passenger cars.

That thirty per cent floor is the right anchor against which to read what is now proposed, because the successor represents escalation rather than adjustment. Under the draft framework, cars below 1000cc would be required to reach eighty per cent domestic value addition by financial year 2031, while sport utility vehicles would face fifty five per cent. Simultaneously, the tariff premium on imported vehicles, reported at around forty per cent, would fall to zero by financial year 2030, and a thirty per cent depreciation cap on imported vehicle pricing has been floated to limit how aggressively used imports can undercut local production.

Figure 5. Localisation targets, old floor against new ambition



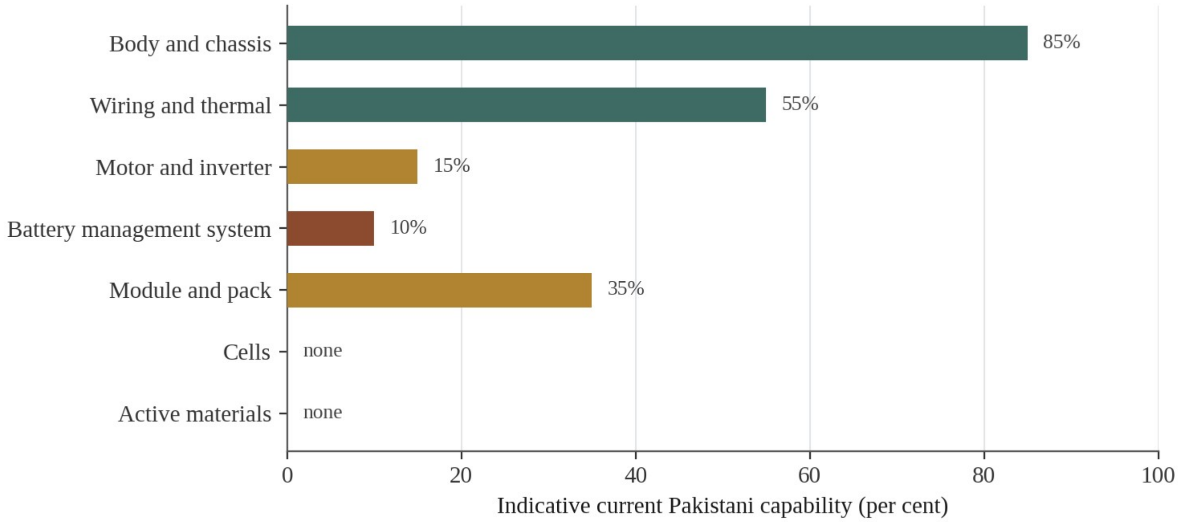
The proposed thresholds represent a near tripling of the old floor, at the same time as tariff protection is scheduled to disappear.

That combination is more demanding than it first appears, and I think its difficulty has been underestimated in most commentary. Raising local content thresholds while simultaneously dismantling the tariff wall means manufacturers must reach a much higher bar at the same time as the protection that made the lower bar survivable is being withdrawn. This is genuinely good policy in the sense that it punishes assembly only operations and rewards manufacturers who build real domestic supply chains, which is what the domestic parts industry has been lobbying for. It is also, on the evidence of the PAVE delivery record examined in section one point two, considerably more ambitious than Pakistan's institutional delivery capacity has recently demonstrated.

As of the writing of this report a steering committee reportedly led by Ishaq Dar had reached in principle consensus on the shape of the successor policy, but it had not moved through the Ministry of Industries and Production's formal process, let alone reached cabinet. In principle consensus is a meaningful signal and more concrete than the working group recommendations behind the battery policy, but it is not notified law, and the specific percentages should be read as the current proposal rather than settled fact. Given how frequently Pakistani automotive policy has been revised between draft and notification, a prudent model should use a range rather than a point estimate, and should watch specifically whether the eighty and fifty five per cent figures survive contact with cabinet review or are softened as comparable targets have been in previous cycles.

For due diligence, the more useful exercise than tracking a headline percentage is disaggregating local content by layer, because a single blended number conceals exactly the substitution this report warns about throughout. Figure 6 sets out the seven layers and my assessment of where Pakistani capability currently sits in each. The point the figure makes is that an eighty per cent headline could in principle be reached almost entirely through body, chassis, wiring and thermal systems, the three layers where Pakistan already has decades of experience, while leaving the motor, the battery management system and the cells themselves fully import dependent. Any assessment of a specific venture's compliance should insist on this breakdown rather than accepting an aggregate figure, because the aggregate tells you almost nothing about whether the strategically important parts of the vehicle are being made in Pakistan.

Figure 6. Local content is not one number, it is seven layers



Analytical assessment based on documented project scope, not a measured production share.

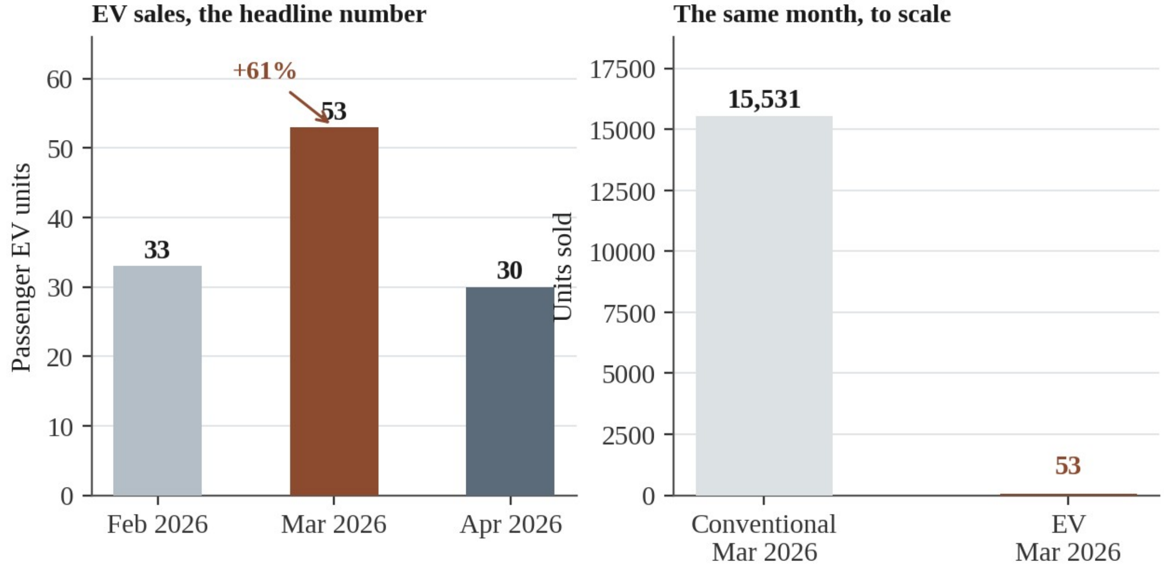
An eighty per cent headline figure can be reached without touching a single strategically significant component.

2. The Market

2.1 The Sixty One Per Cent Jump, Examined

The most widely circulated statistic about Pakistan's electric vehicle market this year is that sales rose sixty one per cent in a single month. The figure is accurate. It originates in Arif Habib Limited's monthly automotive tracking and was subsequently reported by ProPakistani and several other outlets. It describes passenger electric vehicle sales moving from thirty three units in February 2026 to fifty three units in March 2026.

Figure 7. The 61 per cent jump, shown against its own base



Both panels show the same March 2026 data. The right panel adds the denominator that most coverage omits.

Fifty three units is not a market. It is a rounding error against the roughly fifteen thousand five hundred conventional cars, light commercial vehicles, vans and jeeps sold in the same month. The percentage is real and the direction is real, but reproducing it without the base gives an investor a materially inflated impression of current demand, and I have seen the figure quoted in at least one investment context as though it described market share growth rather than twenty additional vehicles.

The surrounding context makes the picture more complicated still. March 2026 was a weak month for conventional vehicles across the board. Passenger car sales fell twelve per cent month on month, with the 1000cc segment, historically the backbone of the Pakistani market, dropping thirty six per cent. Cars below 1000cc slipped eighteen per cent and even the 1300cc and above category declined six per cent. Against that backdrop, a small absolute increase in a statistically negligible segment looks less like electric vehicles pulling ahead on their merits and more like one soft month for the broader market coinciding with modest movement in a category too small to draw conclusions from. It is worth noting that the same month showed a forty per cent year on year increase for the overall market, so the monthly softness sat inside a year of genuine recovery.

April made the trend claim harder still. Model level data showed the Honri Ve, one of the more visible electric models in the reporting, falling forty three per cent month on month from fifty three units to thirty. Whether this represents a genuine reversal, ordinary volatility in a market where a single fleet order swings the percentage dramatically, or a reporting artefact specific to one model cannot be determined from public data. The defensible conclusion is that March confirmed a real demand signal exists and that the public dataset is too fragmented to support any claim about trajectory beyond that.

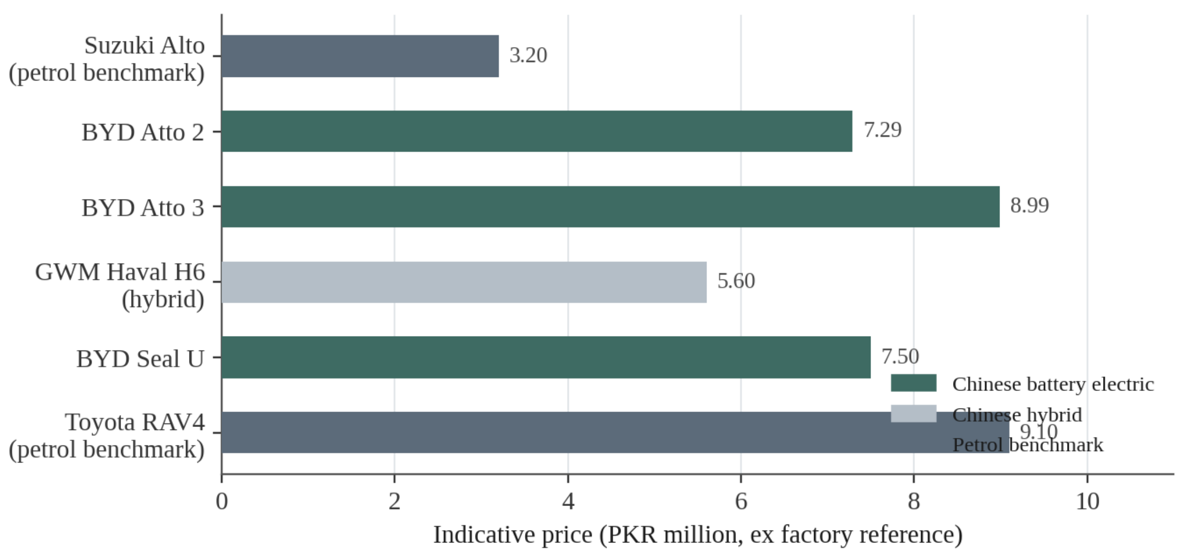
The fragmentation is structural rather than temporary, and it is worth understanding because it affects every number in this section. Pakistan has no single centralised dataset combining monthly electric, hybrid and plug in hybrid registrations by brand and province. The Pakistan Automotive Manufacturers Association bulletins, the most cited source, are built around member manufacturers reporting largely conventional vehicles, with electric figures appearing inconsistently and sometimes aggregated in ways that defeat brand level comparison across months. Provincial registration authorities keep their own records in formats that resist aggregation. Customs data captures imported units but is not published broken down by powertrain in a form that reconciles with either. Anyone building a national picture, including this organisation, has to triangulate across manufacturer disclosures, provincial registrations, customs statistics and company announcements, each capturing a different and partially overlapping slice.

Against the enthusiasm the March figure generated, one countervailing data point deserves attention because it speaks to whether early adoption is converting into durable confidence. Evidence from the 2025 to 2026 period suggests some electric models are losing between fifteen and twenty five per cent of their value within eighteen months, driven by buyer scepticism about battery degradation and range anxiety in a market with no mature secondary market for electric vehicles and no standardised battery health certification. Pakistani car buying culture treats a vehicle as a liquid financial asset with an established resale market, an expectation electric vehicles do not yet meet. This is a genuine demand side headwind sitting alongside the more commonly discussed constraints of price and charging access, and it receives less attention than it deserves.

2.2 Where Chinese Vehicles Actually Sit on the Price Ladder

The most useful reference point for understanding Pakistani vehicle pricing is the Suzuki Alto, which has defined the mass market for decades and continues to do so, moving over seven thousand units in several recent months and taking close to a third of the passenger car market by volume in some periods. No electric vehicle currently sold in Pakistan competes with the Alto on price, and none is likely to at current battery costs. That gap is the clearest available evidence that affordability rather than technology preference remains the binding constraint on mass adoption.

Figure 8. Where Chinese electric vehicles actually sit on the price ladder



Several Chinese electric models now undercut the Japanese petrol benchmark, but nothing approaches the segment that actually drives volume.

Above the entry tier the ladder has widened genuinely over the past year, from what was effectively a single premium niche into a structured set of segments. BYD's Atto 2 launched in late January 2026 at roughly seventy two point nine lakh rupees, making it the most affordable brand new electric sport utility vehicle on sale and positioning it against petrol powered B segment vehicles such as the Chery Tiggo 4 and MG ZS rather than against other electric vehicles. That targeting choice is strategically significant. Rather than competing for share within the tiny existing electric buyer pool, the Atto 2 is aimed at conquest sales from petrol buyers who had not previously considered electric at all.

The Atto 3 sits roughly seventeen lakh rupees above it at approximately eighty nine point nine lakh rupees, offering a larger footprint and a cabin that reviewers and owners consistently describe as unusually premium for the bracket, built on the same blade battery architecture BYD uses across its range. Above that, the Seal U occupies the mid to upper segment at roughly twenty five to twenty eight thousand United States dollars, while GWM's Haval H6, sold as a hybrid in most Pakistani configurations, sits below at eighteen to twenty two thousand dollars. Toyota's RAV4, the benchmark against which Pakistani buyers implicitly compare these vehicles, sits at thirty to thirty five thousand dollars, meaning several higher volume Chinese offerings now undercut the equivalent Japanese hybrid on sticker price, which is genuinely new in a market Japanese brands have dominated for decades.

What this widening implies for localisation is fairly direct. High volume, more affordable models are the plausible candidates for genuine scale driven local content, because their volumes can justify the fixed costs of establishing domestic supply for seats, wiring harnesses and body panels. Premium vehicles will likely remain import heavy regardless of what local content thresholds eventually require, because the volumes rarely justify dedicated tooling and manufacturers at that end will accept a tariff penalty rather than invest in low volume local production. This bifurcation appears in virtually every developing automotive market and there is no reason to expect Pakistan to be an exception.

One further dynamic complicates any simple story about Chinese vehicles undercutting on cost. Every price above is an ex factory or listed reference point excluding freight, registration, withholding tax and dealer documentation charges, all of which add meaningfully to what a buyer actually pays. Because Pakistani vehicle taxation is structured progressively with respect to value, this wedge widens as prices rise, so the effective landed price diverges from the headline figure most at exactly the end of the market where Chinese manufacturers have been positioning their more profitable models. Any competitive benchmarking that uses ex factory prices alone will systematically overstate how competitively these vehicles are priced by the time they reach a showroom floor.

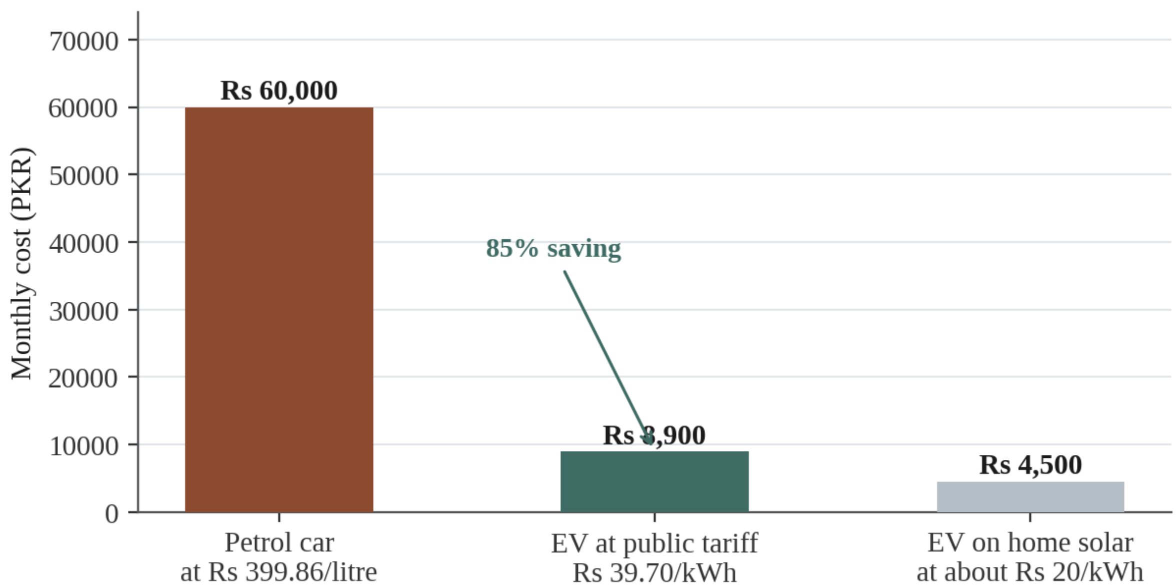
It is also worth noting what the price ladder does not yet contain, because the absence is strategically significant. There is no locally assembled electric vehicle at a genuinely mass market price point, and there will not be until the Gharo plant discussed in section four reaches volume and passes some of its cost advantage through. The most consequential pricing event in this market over the next eighteen months is therefore not a new model launch but the first locally assembled BYD reaching dealers, because that is the point at which the import duty component of the price comes out and the ladder's bottom rung has a chance of moving down toward where Pakistani buyers actually are.

2.3 The Running Cost Arithmetic, Done Properly

A specific comparison has circulated widely enough in Pakistani marketing and social media to have taken on a life of its own, claiming that charging an electric vehicle costs around four thousand rupees a month against a petrol equivalent of roughly sixty five thousand. The comparison is not fabricated. It rests on a single reported case involving a Lahore commuter. It is not, however, a methodology, and it should not be treated as representative.

A defensible calculation needs four published inputs. Petrol currently sells at approximately three hundred and ninety nine rupees and eighty six paisa per litre. A reasonably efficient petrol car achieves around ten kilometres per litre in Pakistani mixed driving. A comparable electric vehicle consumes approximately fifteen kilowatt hours per hundred kilometres, consistent with the compact and midsize electric sport utility vehicles actually sold here. The official electric vehicle charging station tariff, set by the National Electric Power Regulatory Authority and the Power Division under a discounted structure specific to public charging, is thirty nine rupees and seventy paisa per kilowatt hour.

Figure 9. Running cost at 1,500 km per month, using published tariffs



At 1,500 km per month using published tariffs, the saving is 85 per cent. The widely quoted figure describes home solar charging, not the public tariff.

Applied to fifteen hundred kilometres a month, a reasonable approximation of an active urban commuter in Lahore or Karachi, the petrol bill comes to roughly sixty thousand rupees. The same distance charged entirely at the official public tariff costs approximately eight thousand nine hundred. A driver charging at home from rooftop solar at something closer to twenty rupees per kilowatt hour would pay around four thousand five hundred, which is almost certainly where the widely quoted four thousand rupee figure originated, as an outcome specific to solar home charging rather than any market average.

That distinction is not a technicality, and it matters more for infrastructure investors than for vehicle buyers. A network operator building public fast charging and billing at the official tariff offers drivers roughly an eighty five per cent saving against petrol per kilometre, which remains a compelling proposition without any home solar advantage. But it is a fundamentally different business, with different revenue per kilowatt hour, than a market where most charging happens at home on solar and public infrastructure exists mainly to serve intercity travel. Given Pakistan's solar boom and the strong overlap between rooftop solar adopters and early electric vehicle buyers, it is plausible that a meaningful share of current charging happens at the cheaper home rate, which would mean the aggregate advantage for today's buyer base sits closer to the dramatic end of the range even though the viral framing overstates it as a universal figure.

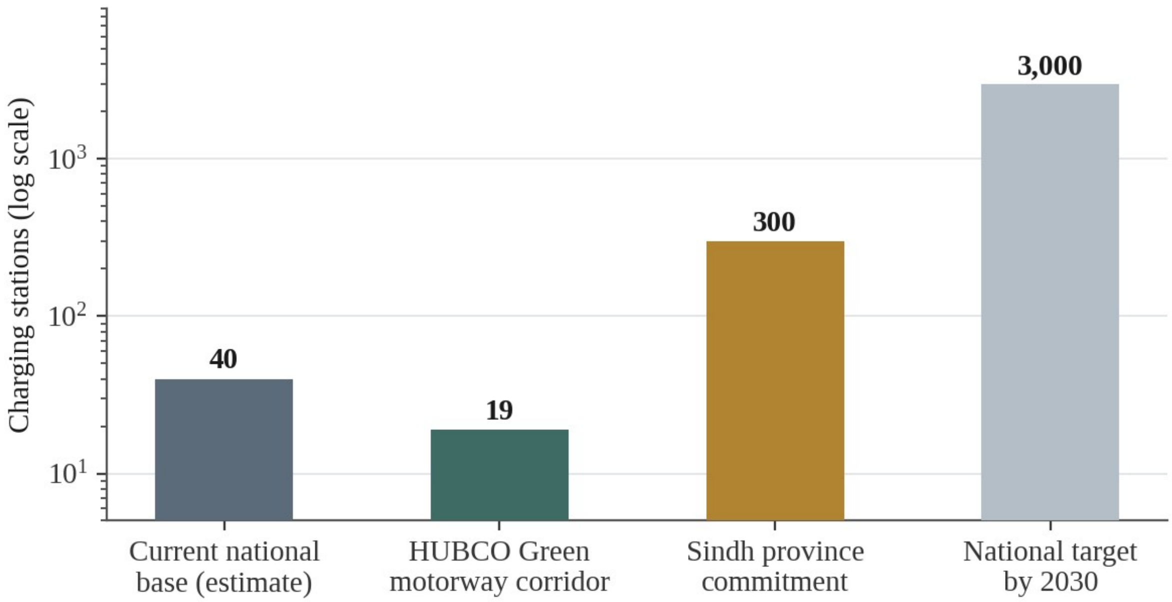
The regulatory history behind the thirty nine rupee seventy paisa tariff is itself worth understanding, because it did not emerge from nowhere and its durability is an open question. The tariff rationalisation process for charging stations, documented in regulator and Power Division filings through 2025 and 2026, involved a deliberate discount against the commercial tariff that would otherwise apply to a business drawing that much power, precisely to make public charging viable for operators while keeping the driver facing price low enough to sustain the value proposition. This is a subsidy in substance if not in name, and subsidies in Pakistan's power sector have a well documented history of coming under fiscal pressure. Any long term model for a charging business should stress test what happens to unit economics if that discount narrows or disappears.

There is a national scale version of this arithmetic that policymakers have used to justify the whole programme, and it is worth recording because it is more specific than most Pakistani policy costings. The government's own analysis projects total electric vehicle energy demand over the five year policy period at one hundred and twenty six terawatt hours, and argues this can be met from existing surplus in the national grid rather than requiring new generation. It further projects a reduction in the fuel import bill from around one hundred and seventy four billion rupees to one hundred and five billion, with carbon credits potentially generating a further fifteen billion in revenue. For the two and three wheeler segment the payback arithmetic is unusually favourable, with an electric bike carrying an additional upfront cost of roughly one hundred and fifty thousand rupees recovering that premium within under two years through fuel savings alone. Whether the surplus generation assumption survives contact with the grid stability concerns discussed in section three point two is a genuine open question, but the costing is at least specific enough to be tested.

2.4 Charging Infrastructure as the Binding Constraint

Pakistan's public charging network is small enough that no officially verified count exists, which is itself informative. The most consistently cited estimate puts the national total between thirty and fifty publicly accessible stations, a range wide enough to reflect genuine uncertainty rather than agreement. Against that base, the national target is three thousand stations by 2030, and the electric vehicle policy specifies a corridor design of roughly forty new stations positioned approximately every one hundred and five kilometres along highways. Sindh has separately committed to a three hundred station provincial buildout.

Figure 10. The charging gap, on a logarithmic scale



A logarithmic scale is necessary because the current base and the 2030 target differ by roughly two orders of magnitude.

The geographic distribution of what exists is described inconsistently by different sources, which again reflects the absence of any authoritative dataset. Islamabad consistently comes out ahead on density, with charging points across Blue Area, Centaurus Mall, and sectors including F six, F eight, I eight, I nine, I ten and G six, alongside the Defence Complex, benefiting from the capital's compact planned layout. Lahore has built what several sources describe as the largest network by station count, spread across Defence Housing Authority areas, Gulberg, Cantt, Johar Town, Bedian Road, MM Alam Road, Thokar Niaz Baig and Packages Mall, with some installations in Defence Phase Six and Gulberg running on solar canopies that keep operating through load shedding, a genuinely valuable resilience feature.

Karachi is where sources actively contradict each other. Some describe it as having the thinnest public network of the three major cities relative to population. Others describe it as having the largest raw count precisely because of the city's size. This is not a discrepancy to wave away. It points to the absence of any single authoritative dataset that different commentators are drawing from, and it means any siting decision premised on Karachi being either underserved or adequately served needs independent verification rather than reliance on published figures.

The development that adds most substance to an otherwise thin picture is HUBCO Green's motorway corridor, which connects directly to the assembly venture discussed in section four. HUBCO Green, established as a wholly owned subsidiary specifically to build a national charging network, has installed nineteen public direct current fast charging stations along the roughly thirteen hundred kilometre Karachi to Peshawar corridor. This is arguably as strategically important to BYD's Pakistani prospects as the assembly plant itself, because a manufacturer can sell electric vehicles indefinitely without expanding the addressable market if buyers remain confined to short urban trips by charging anxiety. A functioning long distance corridor, even a sparse one, changes the calculation for a buyer considering an electric vehicle as a primary rather than a second car.

That the corridor investment comes from the same corporate group building the assembly capacity rather than an unrelated third party is the detail worth dwelling on. It suggests at least one major investor in Pakistan has internalised the lesson that vehicle sales and charging infrastructure have to be developed together rather than sequentially, and it creates an integrated position that a competitor entering later would find expensive to replicate. For a Chinese investor evaluating entry, this is a competitive fact rather than merely an infrastructure fact: the most attractive charging corridor in the country is already controlled by a group with a specific vehicle brand relationship.

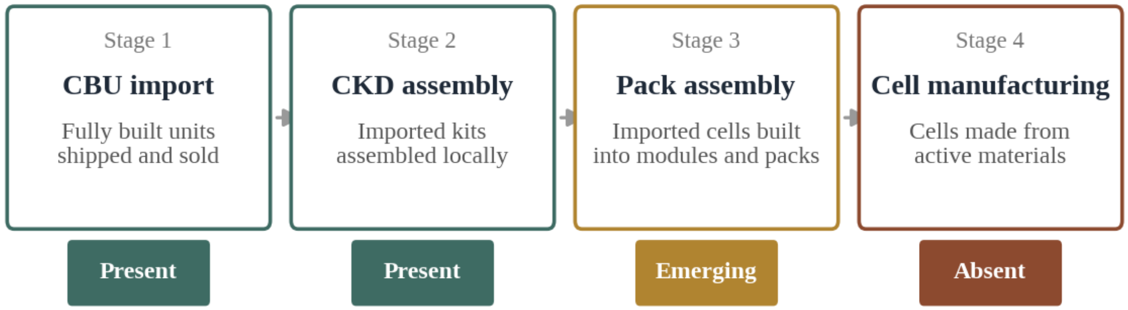
The more useful question than how many chargers exist nationally is how many exist at reasonable intervals along the corridors Pakistanis actually travel, and in the secondary cities outside the well covered triangle. A fourth charger within an already served Lahore neighbourhood adds little. One installed at a sensible interval on the Karachi to Hyderabad stretch, or in Faisalabad, Multan or Peshawar, does considerably more to make electric vehicle ownership practical rather than an urban novelty. Given that the three thousand station target implies an enormous buildout against today's base, the sequencing of that buildout will matter as much as the total, and on the evidence of the delivery record examined in section one point two, the total itself should be treated as directional rather than as a plan.

3. Supply Chain and Mineral Reality

3.1 Assembly, Manufacturing, and the Test That Separates Them

The single most useful discipline an investor can apply to any Pakistani battery or vehicle project is a classification test that takes about thirty seconds and eliminates most of the confusion in this sector. If a project imports finished cells and its Pakistani operation builds them into modules and packs, it is assembly and value addition, which is genuine, useful, investable industrial activity, but is categorically not cell manufacturing. If a project makes electrochemical cells from active materials with real local process control over that chemistry, it is manufacturing in the sense the word carries when governments promise battery industries.

Figure 11. Pakistan's position on the battery value chain



Applied honestly, the test places essentially all current Pakistani activity in the first two stages.

Applied across the current landscape, the evidence overwhelmingly favours the earlier stages. Pakistan has meaningful and growing evidence of pack and system assembly, battery management system integration, imported cells being packaged into finished products, and a rapidly expanding storage distribution network. It has essentially no evidence of cell production. The gaps that would need closing to get there, meaning cell manufacturing capability itself, battery management system design, testing and qualification infrastructure, a functioning recycling loop and upstream material processing, are identified as gaps in the government's own investment promotion material, which is a candid admission worth taking seriously rather than dismissing as boilerplate.

The clearest illustration is the country's first dedicated lithium battery production facility, in the Korangi Industrial Area of Karachi, developed by the Pakistan Solar Association in coordination with the Engineering Development Board. Its significance is real. It is the first facility of its kind, it represents actual capital deployed into battery production rather than vehicle assembly, and it targets the electric two and three wheeler segment, which is where Pakistan has the clearest near term path to meaningful electrification given the price sensitivity documented in section two.

Its limitations are equally real and rarely stated alongside the enthusiasm. Reporting describes an initial capacity around four megawatts, and separately an output of roughly two thousand units a month for small vehicle batteries. Both describe a modest operation by any international standard for battery manufacturing, let alone anything resembling the gigafactory language occasionally used in Pakistani coverage. The facility matters as a first mover and a proof of concept. It does not establish a scaled domestic cell industry, and treating it as evidence that Pakistan has crossed into cell manufacturing would substantially overstate what has been built.

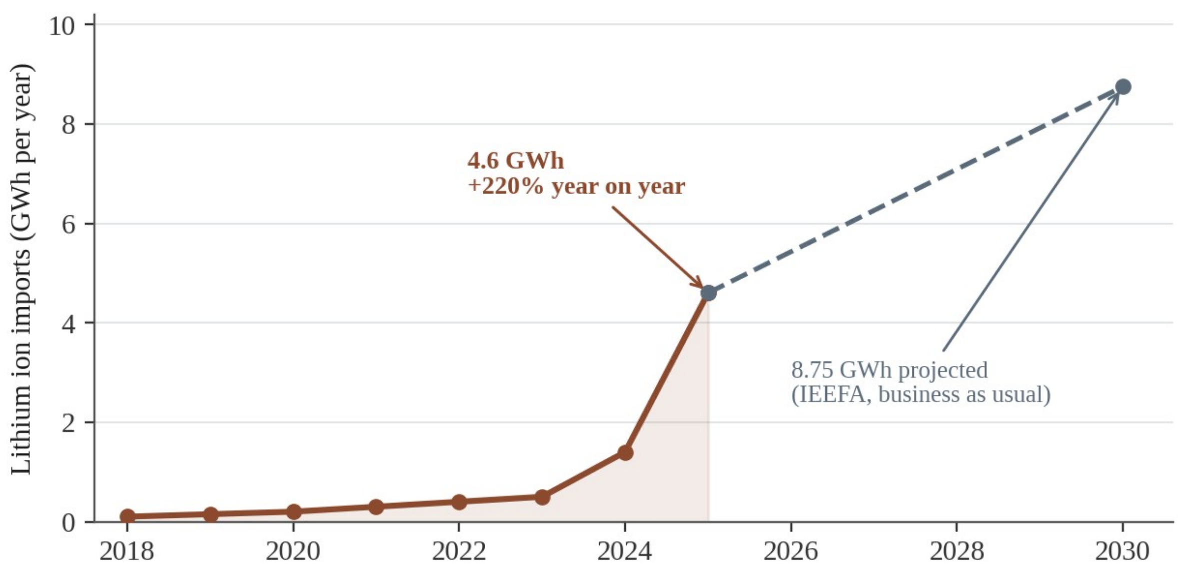
The same terminological slippage runs through the vehicle side. Industry analysts covering the sector have been explicit that under current arrangements Chinese companies are largely importing vehicles and assembling them locally through completely built or semi assembled routes without major investment in localisation or technology transfer. This is not a criticism specific to any manufacturer, and it reflects a rational calculation in a market where demand certainty is limited and the tariff framework is unsettled. But it does mean a facility publicly described as a manufacturing plant may be performing an assembly function adding considerably less domestic value than the word implies, and due diligence should insist on the actual bill of materials and the share sourced domestically rather than accepting a project's self description.

The practical structuring implication is to stage a project's ambitions against the ladder rather than pricing an entire investment as though it will occupy the top rung immediately. A first stage can legitimately and profitably combine imported cells with local module and pack assembly, battery management system integration and testing, all genuine value add over pure import and consistent with actual Pakistani capability. A second stage should target local electrode and material processing. A third, conditioned on cell line economics, qualification pathways once standards are finalised, and a genuine recycling loop, would be the point at which an operation could honestly describe itself as cell manufacturing. A project priced at stage three valuations while operating at stage one should not receive stage three policy treatment or investor confidence.

3.2 Import Dependency and What the Trade Data Shows

The scale of Pakistani dependence on imported cells and packs is best understood through the lithium ion storage import data compiled by Renewables First, an Islamabad think tank whose report From Solar Panels to Storage provides the most detailed public picture available. Cumulative lithium ion battery imports since 2018 reached seven point six gigawatt hours by the end of 2025. The trajectory behind that total is what makes it striking rather than merely large.

Figure 12. A market that switched on almost overnight



Nearly sixty per cent of everything imported since 2018 arrived in a single calendar year.

Annual imports ran between one tenth and half a gigawatt hour from 2018 through 2023, essentially a niche. In 2024 that jumped to one point four gigawatt hours. In 2025 it nearly tripled again to four point six, a two hundred and twenty per cent year on year increase. Nearly sixty per cent of everything Pakistan has imported since 2018 arrived in that single year, a pattern one of the report's authors described accurately as a market that switched on almost overnight rather than one that built gradually. Early indications suggest the pattern held through the first half of 2026, with March and April running particularly high.

A separate projection from the Institute for Energy Economics and Financial Analysis put 2024 battery pack imports specifically at approximately one point two five gigawatt hours, with a further four hundred megawatt hours in the first two months of 2025, and projected imports reaching eight point seven five gigawatt hours by 2030 on a business as usual basis, equivalent to roughly twenty six per cent of projected peak electricity demand at that point. The same analysis flagged a grid stability concern that deserves more attention than it receives in commercial coverage. Unmanaged storage growth at this scale could destabilise the national grid by reducing demand for centrally generated power and raising capacity payment costs, potentially stranding peak generation assets financed on the assumption of a demand profile that behind the meter storage is eroding.

That concern matters for how an investor should read Pakistani policy intent. The growth in battery imports is not simply a demand signal the government wants to accelerate without qualification. It is a trend Pakistan's own energy regulators are simultaneously trying to manage, given knock on effects for a power sector already carrying circular debt and stranded capacity obligations from earlier generation investment. Anyone assuming that policy will move unambiguously toward maximising battery adoption is missing a genuine internal tension within the government's own position, and that tension is likeliest to surface as pressure on exactly the charging tariff discount discussed in section two point three.

This import growth sits directly alongside Pakistan's rooftop solar boom, which is its principal driver. Cumulative installed solar photovoltaic capacity has grown from under two gigawatts three years ago to approximately thirty eight gigawatts by financial year 2025, one of the fastest expansions of distributed solar anywhere in the world over that period. The Federal Minister for Power has unveiled a roadmap targeting ninety per cent clean electricity by 2035, up from roughly fifty five per cent currently, a target implicitly assuming continued rapid growth in both solar and the storage needed to make it dispatchable. The government has also launched an initiative aimed at building local manufacturing bases for clean energy components, an explicit acknowledgment that importing the hardware behind this transition indefinitely represents both a missed opportunity and a growing balance of payments exposure.

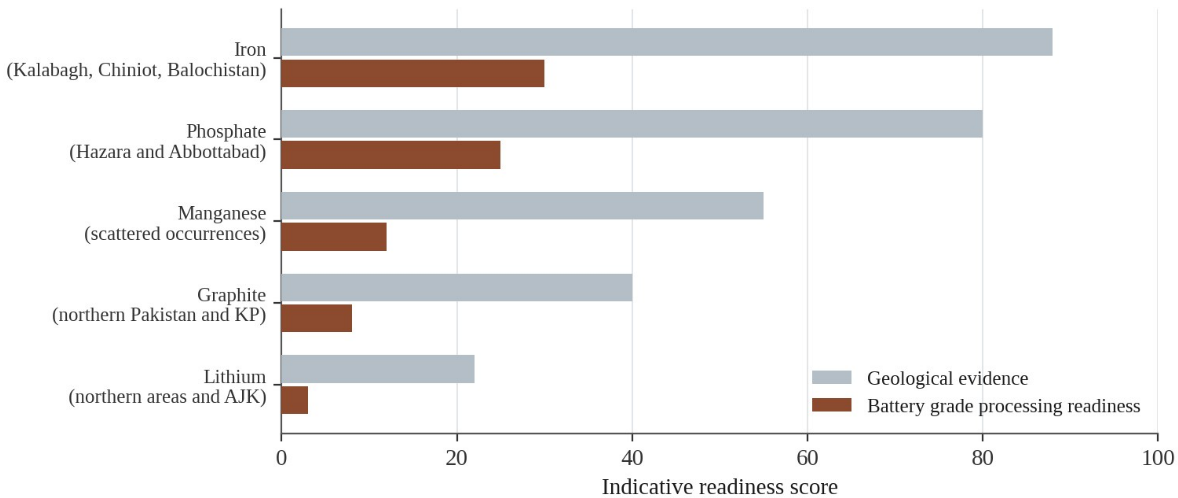
The pricing data attached to this surge explains why local manufacturing would represent a genuine cost advantage rather than merely a nationalist preference. Average lithium ion pack prices in Pakistan range between two hundred and thirty and three hundred and sixty dollars per kilowatt hour, considerably above prices in mature manufacturing markets, a gap attributed primarily to import taxes and customs duties rather than any underlying difference in global cell costs. This produces a genuine policy paradox worth sitting with. The tariff structure meant to protect and incentivise future domestic manufacturing is, in the interim, keeping consumer facing prices elevated in a way that slows the adoption curve the wider policy is trying to accelerate. Resolving that tension is one of the real design challenges facing the framework discussed in section one, and it is not obviously soluble because the two objectives pull directly against each other in the short run.

For anyone building an import substitution thesis, the disciplined approach is to disaggregate rather than treat battery imports as one line. Finished packs and complete storage systems are where local assembly can capture value in the near term, closest to demonstrated capability. Modules sit one step up. Cells, cathode active material, anode material and electrolyte and separator components are progressively harder, requiring process engineering capability Pakistan does not currently possess anywhere in its industrial base. A thesis built on pack and module substitution is grounded in demonstrated capability against a large and rapidly growing addressable volume. A thesis built on cell or cathode substitution is a fundamentally different proposition requiring capability developed from a standing start, and folding the two into one growth narrative is precisely the error this report exists to correct.

3.3 The Mineral Base: Geology Against Battery Grade Reality

Pakistani officials have repeatedly cited domestic phosphate, iron and manganese deposits as the rationale for prioritising lithium iron phosphate chemistry, on the logic that Pakistan possesses a real base for the two elements giving the chemistry its name in a way it does not for nickel or cobalt. The claim deserves to be taken seriously rather than dismissed, because at the level of geological occurrence it is substantially true. It also deserves considerably more scrutiny than it has received, because geological occurrence and battery grade processed material are separated by an enormous amount of metallurgical capability that Pakistan has not demonstrated at scale.

Figure 13. The gap between having the rock and having the material



Analytical scoring based on documented survey evidence, not proven reserve tonnage.

The rock is real. The processing capability that turns rock into battery grade material largely is not.

Working through the inputs individually, iron resources are genuinely well established, with significant deposits documented in the Mianwali and Kalabagh areas of Punjab, at Chiniot, and across parts of Balochistan. Phosphate is similarly well established as a mining category, with an active base centred on the Hazara and Abbottabad districts including newer occurrences identified in more recent surveys. Manganese has documented occurrences across the country, though locations and reserve estimates are less consistently reported than for iron or phosphate. Lithium remains at exploration stage in the northern areas and Azad Jammu and Kashmir, with no proven commercial supply anywhere. Graphite, relevant to the anode rather than the cathode, has documented occurrences in northern Pakistan and Khyber Pakhtunkhwa with quality and scale constraints noted in the survey material.

The distinction running through all five is between a documented occurrence and a proven, commercially viable, battery grade material, and government material citing the mineral base tends to elide it. An iron deposit entirely suitable for conventional steelmaking may require a different and considerably more demanding purification process to reach battery grade cathode purity, where trace impurities acceptable in structural steel meaningfully degrade cell performance and safety. The same applies with greater force to phosphate, where the Hazara and Abbottabad base has been developed and optimised for fertiliser grade output over decades. Converting that supply chain, or building a parallel one, to battery grade purity is a substantial engineering and capital undertaking that has not, on the evidence available, begun at meaningful scale.

For lithium the gap is starker still. Exploration stage resources by definition have not been proven at a scale or grade supporting commercial extraction, let alone battery grade refining. Pakistan should not be counted as having a domestic lithium supply in any near term sense regardless of how the mineral base narrative is framed in policy documents, and an investor who reads the official framing as implying otherwise is being misled by omission rather than assertion.

This matters enormously for how the entire localisation strategy should be read. The rationale for choosing this chemistry rests substantially on the claim that Pakistan has the raw material base to support it domestically. That claim is true at the geological level for iron and phosphate, weaker for manganese, and essentially untrue for lithium and graphite in any near term commercial sense. A genuinely domestic supply chain sourcing battery grade iron and phosphate from Pakistani mines remains a plausible medium to long term outcome given the underlying geology, but it is not a current reality, and no public investment material or survey document this report could access establishes that battery grade processing of Pakistani iron or phosphate has been proven, let alone scaled. This is precisely the kind of open item a Chinese investor should insist on verifying independently through a specific geological and metallurgical assessment against battery grade specifications, rather than accepting the general narrative from ministerial statements.

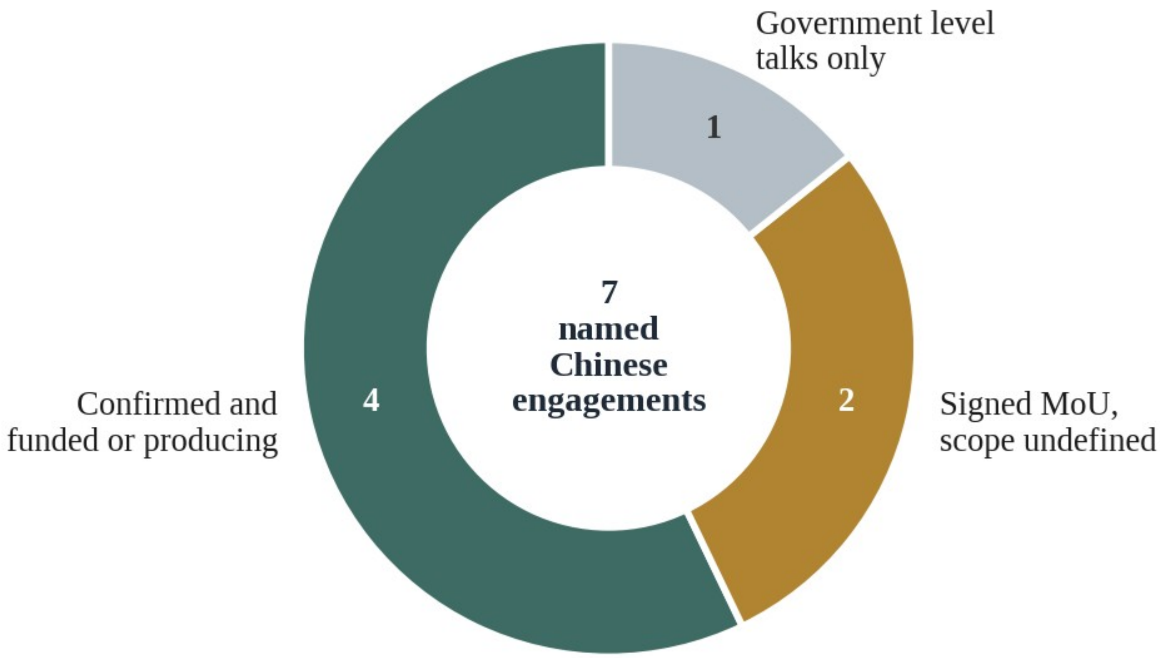
There is a comparative lesson here that Pakistani policymakers and the investors evaluating their claims would do well to keep in view. Indonesia's nickel processing buildout, frequently cited as the template for how a resource endowed developing country forces its way up the battery chain, took the better part of a decade, required tens of billions in smelting and refining capital supplied predominantly by Chinese partners rather than domestic capital, and depended on an ore export ban that left foreign processors no alternative to building onshore. Pakistan has neither the scale of proven reserves Indonesia's nickel base represented, nor an equivalent export restriction mechanism, nor, on current evidence, committed capital approaching what that buildout required. The underlying iron and phosphate geology is genuinely real and genuinely relevant to this specific chemistry in a way few developing markets can claim. The timeline for converting it into a functioning battery grade supply chain is simply far longer and more capital intensive than the current policy narrative suggests.

4. The Chinese Investment Tracker

4.1 Named Engagements, Sorted by What They Actually Commit

The most advanced and best documented Chinese linked investment in the sector is the joint venture between Hub Power Holdings Limited, a wholly owned subsidiary of HUBCO, Pakistan's largest independent power producer, and Mega Conglomerate Private Limited, itself HUBCO's largest single shareholder at roughly nineteen and a half per cent. The vehicle is Mega Motor Company Private Limited, split exactly fifty fifty between the two partners.

Figure 14. The Chinese deal pipeline by confidence level



Four confirmed ventures, two memoranda with undefined scope, and one government level conversation.

BYD's actual role in that structure is frequently misdescribed and is worth stating precisely. BYD has not taken an equity stake in Mega Motor Company. Its involvement runs through a master supply and manufacture agreement, a technical licence agreement and a distribution arrangement, giving it control over technology, branding and component supply without carrying direct equity or balance sheet exposure to the Pakistani venture. This is a common structure for Chinese automakers entering markets where they want a manufacturing foothold without committing capital directly, and understanding it matters for locating where the financial risk actually sits, which is overwhelmingly with the two Pakistani partners rather than with BYD.

The project is a completely knocked down assembly plant at Gharo in Sindh, with a reported cost of approximately one hundred and fifty million United States dollars on a sixty forty debt to equity basis, designed for twenty five thousand vehicles a year. Financial close was achieved in January 2026. HUBCO's own disclosures put its investment in the venture at approximately eight point eight billion rupees as of the end of December 2025, with commercial operations expected in the second half of 2026 and the first locally assembled vehicles reaching dealers before year end. Both foreign and local lenders participated in the debt financing with a term sheet signed ahead of close, indicating conventional project finance discipline and independent lender diligence rather than a purely strategic commitment. The credit rating agency assessing HUBCO has separately noted that the planned electric vehicle investments are expected to materially increase group leverage, which is a candid acknowledgment that this is a genuine balance sheet commitment rather than a costless option.

What makes this a useful test case is that it is, by its own description, assembly rather than manufacturing. Twenty five thousand units a year is meaningful for the Pakistani market, roughly a month or two of total national passenger vehicle sales at current run rates, but volume says nothing about how much value is created domestically. The genuine test will come when the Gharo line begins commercial production and the actual local content of vehicles rolling off it can be assessed against whatever requirements are in force at that point.

On the battery side, Treet Battery Limited signed a memorandum of understanding on 25 May 2026 with Contemporary Nebula Technology Energy Company, in which CATL holds a strategic investment. This should be understood precisely for what it is. It is a real, publicly disclosed agreement between two named entities, establishing a framework to explore energy storage cooperation and potential introduction of CATL cell based products into Pakistan, leveraging Treet's existing distribution and after sales network. It is not a commitment to cell production, and it should not be classified alongside the Gharo project as manufacturing investment. It is a confirmed commercial relationship and a credible pathway toward CATL linked technology, but its own language stops well short of committing either party to local assembly.

Direct engagement between the Government of Pakistan and CATL itself sits earlier still. In May 2026 CATL's Executive President Oscar Luo met Prime Minister Shehbaz Sharif in Hangzhou, with discussions covering advanced battery technology, storage solutions and investment across renewable energy and electric mobility. Pakistan's ambassador to Beijing separately confirmed active discussions, framing them within the second phase of the China Pakistan Economic Corridor, which has shifted emphasis from power generation and infrastructure toward manufacturing and technology transfer. No public reporting discloses an investment figure, plant capacity or implementation timetable. This is a talks stage engagement, significant as a signal of top level interest from the world's largest battery manufacturer, but categorically different from either the Gharo project or the Treet memorandum.

Completing the tracker, a memorandum between StarCharge Energy Pakistan and Bahum Global announced in July 2026 covers charging ecosystem development alongside local manufacturing and technology transfer ambitions, with scope and timeline still to be defined. On the vehicle side beyond BYD, Chery's venture with Master Group has moved furthest, with locally assembled Tiggo 8 and Tiggo 9 models already launched and on sale, making it the most advanced actual production among Chinese branded operations in Pakistan. Jetour has entered through United Motors, and Chery's Jaecoo and Omoda export brands are being developed through NexGen Auto, connected to the Nishat Group. Each should be read through the same lens applied to Gharo: genuine and welcome investment in assembly capacity, not manufacturing in the fuller sense.

4.2 Structuring an Entry Against What This Evidence Shows

For an investor evaluating fresh entry rather than observing existing deals, the accumulated evidence points toward several structuring principles beyond the staged approach already set out in section one. The first is to separate the equity structure from the technology relationship, following the precedent BYD itself established. Providing technology, branding and component supply through licensing and supply agreements rather than direct equity allows participation in Pakistani market growth while limiting balance sheet exposure to currency volatility, an unsettled tariff framework and the broader macroeconomic and political risk any investor here must price.

That is not a universal recommendation. A manufacturer with genuine ambitions to build deep local capability rather than simply access the market through a licensed partner may reasonably prefer direct equity to maintain control over execution and quality. But for an investor whose primary interest is market access and brand presence, the licensing structure deserves serious consideration as the lower risk entry, and the fact that the most sophisticated Chinese entrant to date chose exactly this route is evidence worth weighing rather than coincidence.

The second principle is to insist on defining measurable technical milestones that separate the value chain stages set out in section three, rather than leaving them as aspirational language. Both the Treet and StarCharge agreements are currently structured as frameworks for exploring cooperation rather than binding commitments with defined deliverables. That is a reasonable first step in an unfamiliar market. But any investor entering a similar framework should push, in subsequent definitive agreements, for specific contractually defined indicators around cell, module and pack level output that allow both parties, and any outside observer conducting later diligence, to assess objectively whether the partnership is progressing up the ladder or remaining static at assembly and distribution indefinitely.

The third concerns how to treat unfinalised policy when negotiating today. Rather than waiting for finalisation, which risks losing first mover positioning, or signing binding long term commitments premised on pitchbook figures as though they were law, the defensible middle path builds explicit renegotiation triggers into definitive agreements, specifying that key terms around tariff treatment, local content obligations and incentive dependent capital commitments will be revisited on formal gazettal if notified terms differ materially from what circulated during negotiation. This is standard practice for industrial investment in jurisdictions with genuinely evolving frameworks, and its absence from an agreement signed in Pakistan today should be read as a signal about the drafting party's sophistication rather than as confidence in the policy trajectory.

The fourth follows from the PAVE record in section one point two and applies specifically to any project whose economics depend on a government disbursement rather than a tariff concession. Tariff concessions operate at the border and require no administrative discretion once notified. Subsidy disbursements require a functioning payment chain, and Pakistan's demonstrated record on that chain is thirteen per cent delivery against a fully funded first phase. Any project relying on subsidy flow rather than duty relief should model a substantial delay assumption and, where possible, negotiate for the tariff based version of an incentive over the disbursement based version even at a lower headline value, because a smaller concession that operates automatically is worth more than a larger one that depends on an administrative process with this track record.

Finally, given the distinction running through this tracker between CATL as a corporate entity and the separate company in which it holds a strategic investment, any investor should be precise in its own communications and internal risk assessments about which relationship it is actually building on. Describing a Treet style partnership as CATL entering Pakistan, when the counterparty is a CATL backed but separate company operating at memorandum stage while CATL itself remains at preliminary government discussions, materially overstates the depth of involvement in ways that could mislead co investors, lenders or partners relying on that characterisation. This kind of imprecision erodes credibility with sophisticated counterparties as soon as the underlying facts are examined, which they invariably are.

5. Pakistan Against Its Regional Competitors

5.1 Bangladesh and the Discipline of a Sunset Clause

Bangladesh's approach, set out in its financial year 2026 to 2027 budget proposals, is built around a design principle Pakistan's framework has not matched in clarity, even though both countries are pursuing broadly similar goals. The core mechanic is a duty structure tiered explicitly by the degree of local value addition a manufacturer actually undertakes, rather than a flat concession available regardless of contribution. Manufacturers carrying out body fabrication, welding, painting and assembly of three and four wheeled electric vehicles with high local value addition receive exemption from essentially all duties and taxes on imported materials and parts, retaining only a nominal three per cent import duty. Manufacturers conducting lower value operations, essentially assembly and painting without fuller fabrication, face fifteen per cent on the same inputs.

That five fold differential between three and fifteen per cent creates a sharper and more legible incentive to move up the value chain than anything currently notified in Pakistan, where the completely knocked down tier system pushes in the same direction without as clean a binary. It is worth noting that Bangladesh achieves this within a single budget instrument, where Pakistan's equivalent is spread across an expired auto policy running on extension, an unfinished battery policy and a separate unfinished battery specific instrument.

On the consumer side, Bangladesh is cutting the total tax burden on imported electric vehicles priced up to twenty five thousand United States dollars from ninety three to sixty four per cent, with vehicles up to fifty thousand dollars facing eighty per cent. The advance income tax on registration, previously a flat and often prohibitive two hundred thousand taka regardless of specification, is being restructured to scale with battery capacity, falling to twenty five thousand taka for two hundred kilowatt vehicles and capped at seventy five thousand for the largest category. Charging infrastructure receives dedicated treatment, with the previously applicable thirty nine point seven five per cent duty and tax burden on imported chargers eliminated entirely, a more aggressive charging incentive than anything currently notified in Pakistan. Simultaneously the tax burden on petrol and diesel cars between 1,200cc and 1,600cc rises from one hundred and thirty two to one hundred and fifty six per cent, pushing from both directions at once.

The feature that most distinguishes the Bangladeshi design, and the one I consider most instructive, is the explicit sunset clause on concessions covering the most strategically important components. Concessionary duty facilities for lithium ion battery manufacturing, sodium ion production and pack assembly have been extended only through financial year 2029 to 2030, while concessions specifically covering battery components and mounting brackets run only to June 2028. This is a deliberate build the ecosystem then remove the crutch design, giving established players a bounded window to establish functional cell and pack production before importing the same components becomes substantially more expensive. It is a clearer structure than an open ended assembly subsidy, and it stands in contrast to developing country incentive schemes that have remained open for decades past the point of serving their original purpose.

It would nonetheless be a mistake to read this more codified structure as evidence that the underlying market is ahead of Pakistan's. Bangladesh Road Transport Authority figures record only six hundred and sixty nine electric vehicles formally registered nationally as of mid May

2026, with formal registration having begun only in September 2022. That is a smaller base than Pakistan's, and it means Bangladesh's well designed architecture is a bet on future growth rather than a reflection of a thriving present. Some Bangladeshi business leaders have themselves warned that the incentive structure may inadvertently favour imports over domestic manufacturing in early implementation, which echoes precisely the assembly against manufacturing distinction this report applies to Pakistan and confirms that a better designed policy on paper does not automatically produce a better outcome in practice.

For a Chinese investor weighing the two, the honest comparison is between a market with a smaller current base but a more legally certain and explicitly tiered structure with a defined sunset, against a market with a comparably small base but a considerably larger population, a longer established assembly tradition through decades of Japanese manufacturer operations, and a framework that remains substantially less codified. Neither offers a mature ecosystem. The choice comes down to which risk an investor is better positioned to manage, policy uncertainty in Pakistan or a genuinely thin current market in Bangladesh, rather than any objective superiority of one over the other.

5.2 Vietnam and the Value of a Codified Framework

Vietnam belongs to a different category of comparison, not because its individual incentives are necessarily more generous line by line, but because it enters with an established domestic manufacturing base and a legal architecture for foreign investment incentives comprehensively overhauled within the past year. The National Assembly passed an entirely new Corporate Income Tax Law on 14 June 2025, replacing the previous framework in full, effective from 1 October 2025 and applicable retroactively to the whole 2025 tax year. This is a complete legal overhaul rather than incremental amendment, and its recency means an investor evaluating Vietnam today works from the current architecture rather than an older framework layered with successive patches.

The regime follows a familiar tax holiday structure applied through more tightly defined criteria than before. Qualifying projects, including high technology application, strategic technology and supporting industry categories that a genuine cell manufacturing or advanced material processing facility would plausibly meet, are broadly eligible for full corporate income tax exemption following the first profitable year of incentivised activity, followed by a period at a fifty per cent reduced rate, with duration varying by category and location. Under the predecessor framework, comparable supporting industry projects received a ten per cent preferential rate for fifteen years combined with four years full exemption and nine further years at half rate, which gives a sense of the order of magnitude involved.

A significant structural change removes what had been an easy qualifying route. Locating within an ordinary industrial zone no longer qualifies a project for incentives on its own. Projects must now qualify through genuine sector criteria demonstrating high technology, digital or green characteristics, or through geographic criteria specific to special economic zones or designated difficult areas. Projects licensed before the new law may continue under previous terms until expiry. This tightening reflects Vietnam's own assessment that blanket industrial zone incentives had become insufficiently targeted, and it is worth noting as evidence of a state actively refining its incentive architecture rather than simply accumulating concessions.

Beyond corporate tax, Vietnam layers additional measures the South Asian markets do not match in combination. Decree 202 of 2026 extends full exemption from the registration fee for battery electric vehicles through 31 December 2030, framed explicitly by Vietnamese officials as providing the policy certainty manufacturers require when committing capital on the five to ten year horizons typical of automotive decisions, which is precisely what this report has identified as missing in Pakistan. Businesses may allocate up to twenty per cent of annual pre tax profit into a tax deductible research and development fund, with qualifying expenditure deductible at up to two hundred per cent of actual cost under a more recent decree, a genuinely generous innovation incentive exceeding anything on offer in either South Asian market.

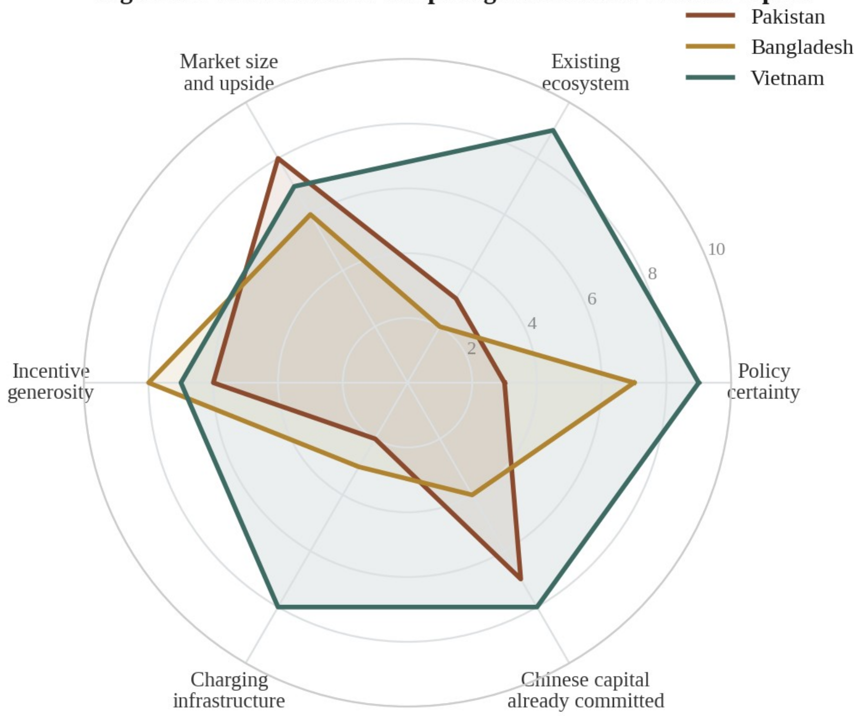
One complication a sophisticated investor must weigh has no equivalent in the Pakistani or Bangladeshi context. Vietnam has implemented the global minimum tax framework through a domestic decree effective from mid October 2025 with retroactive application to financial year 2024, applying to multinational groups with consolidated global revenue of at least seven hundred and fifty million euros in two of the preceding four years. Both CATL and BYD comfortably clear that threshold. For groups of that scale, the fifteen per cent effective minimum enforced at consolidated level can claw back much of the benefit of a zero or low preferential rate applied at the Vietnamese subsidiary level. This is technical but material, and it means Vietnam's headline incentive package is worth considerably less to exactly the largest Chinese battery manufacturers most likely to be considering a genuine cell investment than it is to smaller investors falling below the threshold.

That asymmetry is worth stating plainly because it partially closes a gap that headline comparison would suggest is unbridgeable. If Vietnam's principal advantage over Pakistan were purely the generosity of its tax holidays, the comparison would be straightforward. Because a substantial portion of that generosity is neutralised by the global minimum tax for the specific investors most relevant to this sector, Vietnam's genuine remaining advantages are its existing ecosystem, its infrastructure and its policy predictability, rather than its net fiscal offer. Those are real advantages, but they are a narrower set than the raw incentive comparison implies.

5.3 What the Comparison Actually Resolves

Bringing the three together, the choice facing an investor is not a ranking exercise with one winner but a genuine trade off between different combinations of certainty, maturity and unrealised upside. Figure 15 sets out my assessment across six dimensions.

Figure 15. Three markets competing for the same Chinese capital



Analytical scorecard, not an official ranking. Scores reflect this report's assessment of the evidence reviewed.

Vietnam leads on almost every dimension except the one that matters most to an early mover: unrealised upside.

Vietnam offers the most legally certain framework, backed by a functioning domestic manufacturing base and a freshly overhauled tax law, but bundled with a more crowded competitive landscape, higher operating costs than either South Asian alternative, and for the largest investors the global minimum tax offset just discussed. Bangladesh offers the most cleverly tiered incentive design of the three, with a defined transition timeline forcing genuine localisation on a schedule rather than allowing indefinite import dependent assembly, but sitting on a market so early stage that any case must rest almost entirely on projection.

Pakistan offers the largest underpenetrated addressable market in raw population terms, a growing base of confirmed Chinese commercial engagement, and demand side fundamentals that are arguably better documented than either alternative's, specifically the running cost arithmetic in section two and the storage import trajectory in section three. What it does not offer is comparable policy certainty. The battery manufacturing policy remains pre cabinet. The battery specific instrument remains unfinished. The successor auto policy remains at steering committee consensus. An investor placing high value on regulatory predictability, with flexibility to deploy elsewhere, would reasonably weight that heavily and might conclude the uncertainty is inadequately compensated.

That is not the only reasonable conclusion from the same evidence, and I would not want to leave the impression that Pakistan should simply be discounted. An investor prioritising early mover positioning, willing to accept policy risk in exchange for a foothold in the least contested of the three markets, has a sound basis for reading the gap between ambition and notified law as the opportunity rather than merely the risk. Only one major Chinese manufacturer has established genuine assembly scale in Pakistan to date. CATL remains at talks stage. The competitive field is thinner than in either alternative market, and it will not stay thin indefinitely once the policy framework settles.

The structure that reconciles these positions is the one recommended throughout this report. Tranche commitments against notified rather than proposed policy. Insist on measurable technical milestones. Build renegotiation triggers into terms signed ahead of gazettal. Prefer tariff based incentives over disbursement based ones given the delivery record. Follow the licensing rather than equity precedent where market access rather than deep capability is the objective. None of these eliminates Pakistani policy risk, and nothing available to a foreign investor could. What they do is convert an unhedged bet on a pitchbook into a staged position where each increment of capital is matched to an increment of legal certainty that has actually arrived, which is the most any investor can reasonably ask of a market at this stage of development.

6. What to Watch, and What It Adds Up To

6.1 The Triggers That Will Resolve the Open Questions

Because so much of this report turns on the distinction between notified and proposed, the most useful practical contribution it can make beyond point in time analysis is specifying exactly what would resolve each open question. Several have defined triggers attached rather than remaining permanently uncertain.

Figure 16. The five triggers worth watching

Four of the five decisive events remain pending. The first is scheduled and will arrive before the rest.

Gazetting of the National Lithium Ion Battery Manufacturing Policy, converting the seventy and eighty per cent targets and the zero per cent cell duty from pitchbook figures into enforceable law, is the single most consequential trigger for the battery investment case. Its timing determines when a staged approach can move from its initial notified tranche into the cell and material processing stages that depend on the finalised schedule. Publication of the finalised battery testing and certification standards is closely related, since these determine the capital expenditure and qualification timeline any manufacturer must budget for, and their absence means even a manufacturer ready to invest cannot yet plan against a known compliance target.

On the vehicle side, formal notification of the successor auto policy, converting the eighty and fifty five per cent value addition proposals from steering committee consensus into gazetted law, is the equivalent trigger. The percentages that survive that process, and whether the reduction of the import tariff premium to zero by financial year 2030 survives intact, will materially affect the economics of every assembly venture currently operating.

Beyond these, the most useful real time indicator of whether localisation is actually occurring rather than merely being promised would be Pakistani customs import data broken down by battery component category, distinguishing cells from modules from finished packs from cathode and anode materials. This granularity does not currently exist in published trade statistics. Establishing it, whether through direct engagement with the Federal Board of Revenue or through modelling from shipment level trade data, would do more than any other single series to separate genuine progress up the value chain from assembly dressed in manufacturing language, and it is a gap this organisation intends to work on.

Two further triggers provide the clearest concrete test of the distinction running through this report. The first is the actual local content composition of vehicles from the Gharo line once commercial production genuinely begins, which offers the first real world test of whether Pakistan's most advanced Chinese linked project is a step toward manufacturing or remains assembly in substance regardless of description. The second is whether the Treet relationship progresses beyond memorandum into a definitive agreement covering actual local assembly or value addition, which would be the first concrete test of whether CATL linked technology is moving toward industrialisation or remaining exploratory. Secondary city and corridor charging deployment outside the three major cities is worth tracking as an independent indicator of whether the market is broadening geographically or remaining concentrated in a narrow urban band.

The tracking architecture this organisation intends to apply is a five panel monthly update covering notified policy changes distinguished clearly from proposed ones, vehicle registrations by brand and powertrain to the extent the fragmented sources allow, battery import volumes by component category, charging locations distinguished between major city concentration and genuine corridor expansion, and joint venture status tracking whether each named engagement is progressing, static or regressing. Every data point should carry a source, a date, a geography and an explicit confidence grade. The single greatest service any ongoing analysis of this sector can provide is simply refusing to blur those categories together.

6.2 The Assessment

Working through the policy architecture, the market data, the supply chain, the mineral endowment, the named tracker and the regional comparison in turn, the conclusion I arrive at is that the demand conditions for a genuine Pakistani battery and electric vehicle industry are real, and on the evidence assembled here they are stronger than the level of international attention this market receives would suggest. Storage imports growing at over two hundred per cent annually, a verified running cost advantage of roughly eighty five per cent against petrol on published tariffs, a large and chronically fuel import dependent economy with every incentive to reduce that dependence, a notified vehicle policy funded by a dedicated levy rather than an annual budget fight, and credible named Chinese engagement spanning assembly, battery memoranda and government to government discussion all point the same way.

What is not there is the industrial depth to convert that demand into a domestic manufacturing base, and it will not arrive quickly regardless of how the pending policies are finalised, given the multi year timelines that cell manufacturing, battery grade mineral processing and genuine technology transfer have required in every comparable market examined here. Pakistan has no cell manufacturing. Its mineral endowment is geologically real and metallurgically unproven. Its charging network is two orders of magnitude below its own stated target. Its most recent fully funded subsidy programme delivered thirteen per cent of its first phase target before being redesigned.

The most investable opportunities available to a Chinese partner today are consequently not the cell gigafactory propositions that occasionally circulate in Pakistani coverage. They are the more modest and more immediately achievable layers: pack and module assembly, battery management system integration, thermal systems, testing and certification infrastructure, recycling, charging network development, and battery grade mineral processing specifically targeted at the iron and phosphate base that is geologically genuine even where commercially unproven. Each of these has a real addressable market today, each is consistent with demonstrated Pakistani capability, and each positions an investor to move up the ladder as the policy framework settles rather than betting on a rung that does not yet exist.

Cell manufacturing remains the strategic prize that both Pakistani policymakers and the Chinese investors evaluating this market are, reasonably, aiming at. Reaching it will require policy certainty Pakistan has not yet delivered, quality standards it has not yet finalised, a power supply reliability it continues to struggle with, committed technical partners beyond the memoranda currently in place, and upstream processing capability it has not yet built. That is a long list. It is not, however, an impossible one, and the direction of travel across every section of this report is toward those conditions rather than away from them. The question for an investor is not whether Pakistan will eventually get there. It is whether the option value of being positioned early, in the least contested of three competing markets, compensates for underwriting a policy framework that is still being written. Reasonable investors will answer that differently, and this report has deliberately avoided pretending there is a single correct answer, because on the evidence available there is not.

Sources and Method

This report draws on government and regulatory sources including the Ministry of Industries and Production, the Engineering Development Board, the Press Information Department, the National Electric Power Regulatory Authority and the Power Division, the Geological Survey of Pakistan, the Punjab Mines and Minerals Department, and the United States Geological Survey. Corporate and financial sources include Pakistan Credit Rating Agency rating reports on Hub Power Company, Hub Power Holdings and Mega Conglomerate, HUBCO corporate disclosures, and Pakistan Stock Exchange disclosures on the Treet Battery agreement. Market data derives from Pakistan Automotive Manufacturers Association monthly bulletins as reported by PakWheels and ProPakistani, and Arif Habib Limited monthly automotive tracking. Analytical and think tank sources include Renewables First, the Institute for Energy Economics and Financial Analysis, the Pakistan Institute of Development Economics, and the Sustainable Development Policy Institute. Regional comparison draws on the Bangladesh financial year 2026 to 2027 budget documentation and Bangladesh Investment Development Authority material, and on Vietnamese Corporate Income Tax Law 67 of 2025 together with implementing decrees. Press coverage from Dawn, Business Recorder, the Express Tribune, Nikkei Asia, Arab News, the Daily Star, the Financial Express and Prothom Alo has been used where primary documentation was unavailable.

Figures marked as analytical scoring represent this report's assessment of the reviewed evidence rather than measured production shares or official rankings, and are identified as such in their captions. Where sources conflict, the conflict is reported rather than resolved silently. Where a figure could not be verified against a primary or credible secondary source, it has been excluded rather than reproduced with a qualifier.

About the Author

Reza Khan is a researcher with an M.Phil. in Government and Public Policy from the National Defence University, Islamabad, and an MBA in Finance. His professional experience spans business strategy, international trade, economics and organizational leadership, and he has taught Development Economics, Microeconomics and Supply Chain Management at the University of London Global MBA programme. His analytical interests include public policy, governance, strategic communication and the interaction between technology, institutions and society.